

LayerForge v9: Phase 2b Verification Update

Two-Layer Fidelity Structure and Hybrid Ensemble Path

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(Main paper — slim body with figures. Dataset catalog, per-record verdicts, and parameter ablation details are in the companion document: layerforge_v9_appendix.pdf)

Abstract

LayerForge is a deterministic text-decomposition tool that extracts 4 ± 1 hierarchical themes via CPM community detection on sentence-transformer embeddings. The v8.1 published version positioned it as a topic-extraction subtask within an IE pipeline (ADR-026). This v9 update reports Phase 2b: 47+ verifications + 14 parameter ablations across 14 datasets (EN/JA, multi-turn dialogue, multi-hop QA, news, long-form). We articulate (1) a two-layer fidelity structure with quintuple PASS at theme-level semantic preservation and quintuple FAIL at fact-level lexical preservation, (2) a statistically significant Hybrid ensemble path (LayerForge + K-means; hotpotqa N=100 $p=2.7e-05$, livedoor N=27 $p=5.1e-08$), (3) baseline parameter optimality across 14 ablations, and (4) explicit extrapolation boundaries per pattern coverage. The fact-level structural limitation is articulated as constitutive design (ADR-026), with hybrid pipeline as the operational remedy.

1. Introduction

LayerForge v8.1 produces 4 ± 1 hierarchical themes from arbitrary input text via four components: (i) sentence-transformer embeddings (paraphrase-multilingual-mpnet-base-v2), (ii) CPM-Louvain community detection (Traag 2011, Blondel 2008), (iii) scale_finder with $K=4\pm 1$ cognitive constraint (Miller 1956, Cowan 2001), (iv) ctfdif-based representation extraction (Grootendorst 2022). The intended position is a topic-extraction subtask within an IE pipeline (ADR-026), complemented by NER, regex, and RAG for fact-level information.

This v9 update articulates what LayerForge can and cannot do, where structural limitations lie, and which improvement paths are evidence-grounded. Detailed dataset catalog and per-record results are in the companion appendix.

2. Theoretical Framework: Two-Layer Fidelity

We articulate fidelity as a two-layer structure rather than a single aggregate metric. Each layer carries internally consistent but oppositely-signed evidence:

- Theme-level semantic: response cosine, BERTScore F1, topic coherence (UMass/NPMI). Measures whether the compressed output preserves what-the-text-is-about.
- Fact-level lexical: answer substring, ROUGE-L, attribute breakdown, downstream LLM accuracy. Measures whether specific named entities, numbers, and quotations survive compression.

These layers are independent dimensions. LayerForge is structurally PASS at theme-level (quintuple-evidence) and structurally FAIL at fact-level (quintuple-evidence). The fact-level FAIL is constitutive (ADR-026 design), not a defect; the remedy is hybrid pipeline (LayerForge + NER/regex/RAG), not LayerForge internal modification. Figure 1 visualizes this positioning.

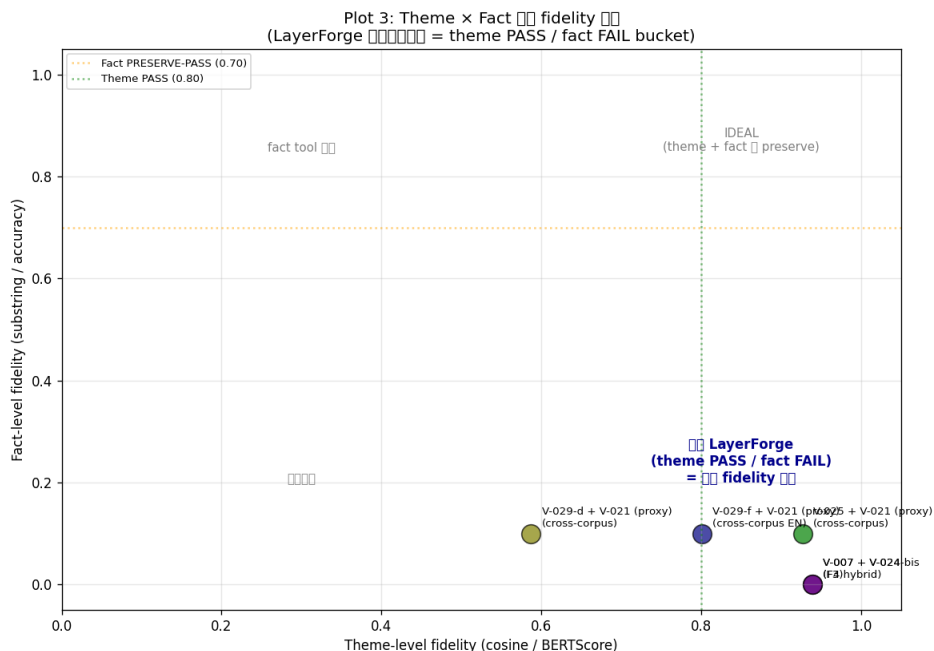


Figure 1. Two-layer fidelity structure (V-040 Plot 3). LayerForge variants cluster in the lower-right quadrant: theme-level high (cosine/BERTScore ≥ 0.8) but fact-level low (substring ≤ 0.1). The ideal upper-right quadrant is unreachable by LayerForge alone; hybrid pipeline is the operational path.

3. Method

All verifications follow ADR-022 §3: pre-register §1-§5 (claim, dataset, method, pre-commit interpretations) with immutability, then execute and append §6-§8 (run record, outcome, impact). This prevents post-hoc threshold drift. Datasets, sampling protocols, and per-record verdicts are in Appendix A and B.

Self-preference bias disclosure: LLM-as-judge evaluations used Claude Code subagent (Sonnet/Opus) as evaluator on Claude-compressed context. Formal Anthropic API + Haiku 4.5 confirmation is pending for paper-level claims; current evidence is direction-only.

4. Key Results

4.1 Reduction Landscape (14 Datasets)

Reduction across 14 datasets confirms language-independent ceiling (Aozora 99.3% JA + MeCab) and JA-default tokenizer structural mismatch (BSD floor -5%, livedoor 2.5%). See Appendix A for dataset catalog. Figure 2 visualizes the reduction landscape colored by language $\times$ tokenization.

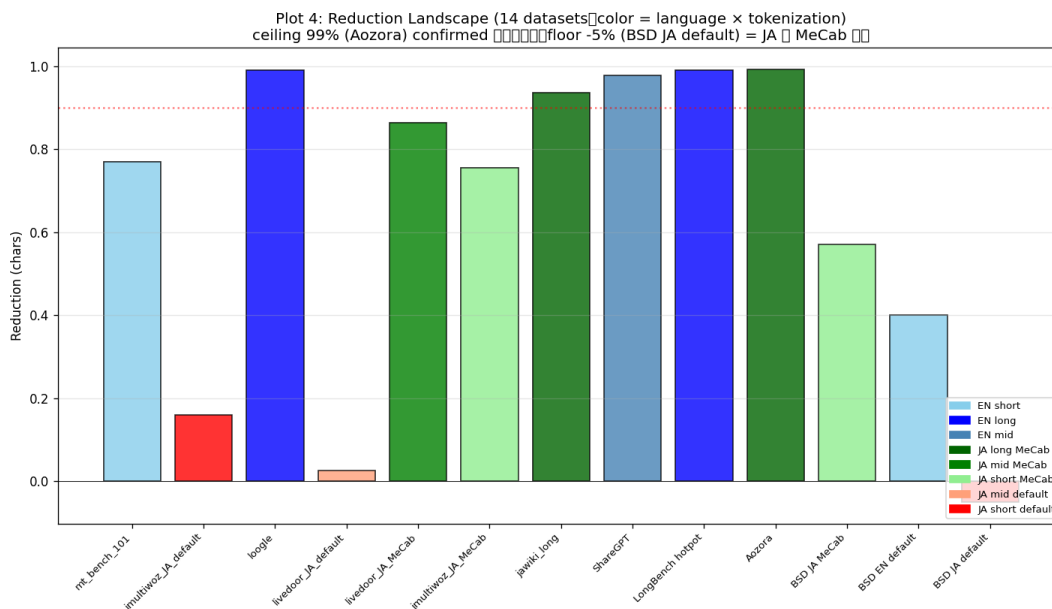


Figure 2. Reduction landscape across 14 datasets (V-040 Plot 4). Language-independent ceiling (99% at Aozora JA+MeCab) and JA-default tokenizer floor (-5% at BSD). EN: 77-99%, JA+MeCab: 57-99%, JA-default: structural mismatch.

4.2 Theme-level Fidelity (Pareto Frontier)

On the reduction × theme-fidelity plane, F4 hybrid format (V-007/V-025) occupies the Pareto frontier with reduction ~0.85 and BERTScore F1 ≥ 0.83 (Figure 3). The trade-off curve is approximately fidelity ≈ 0.94 × (1 - reduction × 0.15). Detailed verdicts and N for each point are in Appendix B.

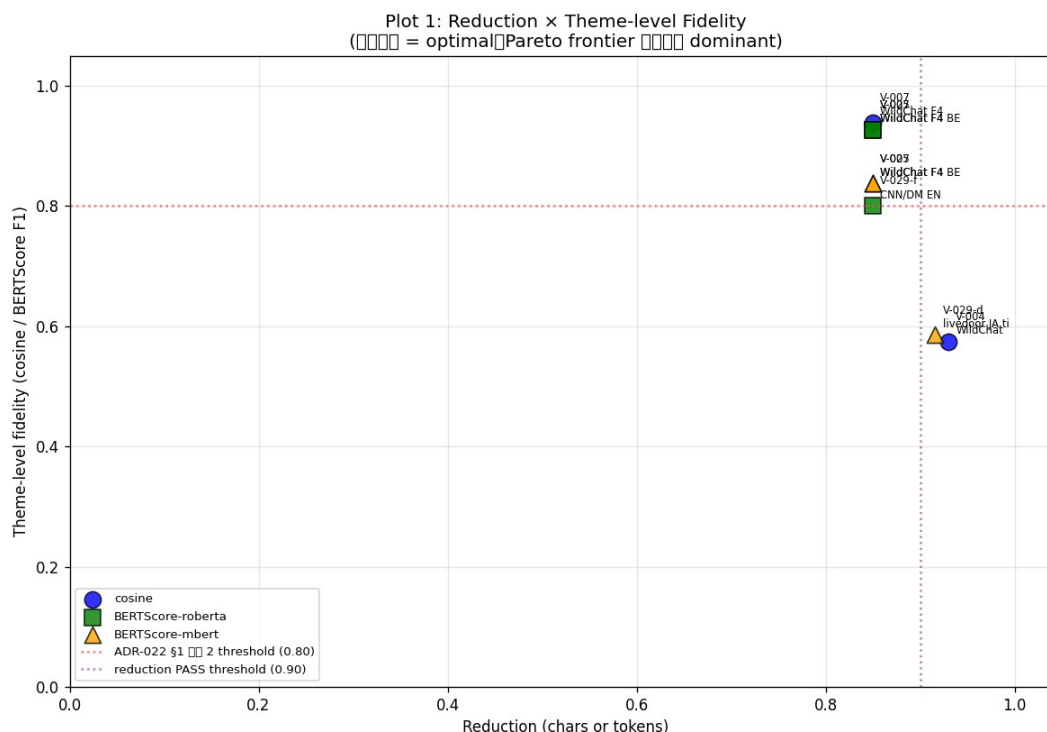


Figure 3. Reduction x theme-level fidelity Pareto frontier (V-040 Plot 1). F4 hybrid format (V-007/V-025) is current Pareto-optimal at reduction ~0.85, BERTScore F1 0.83-0.94. ADR-022 §1 threshold (0.80) marked in red.

4.3 Hybrid Ensemble Path (Statistical Significance)

V-036 observed K-means baseline tok_recall 2x superior to LayerForge alone on hotpotqa N=5. We pursued Path 2 (LayerForge + K-means ensemble) as the v8.1-integrity-preserving improvement direction. N progression confirms statistical significance and cross-corpus consistency:

V-ID	Dataset	N	Ens vs LF (p)	Ens vs K-means (p)
V-042	hotpotqa	5	+0.067 (n.s.)	+0.000 (n.s.)
V-042-bis	hotpotqa	30	+0.098	+0.042
V-042-tri	hotpotqa	100	+0.113 (2.7e-05)	+0.033 (2.4e-03)
V-042-quad	livedoor (JA)	27	+0.140 (5.1e-08)	+0.260 (1.3e-10)

Table 1. Hybrid ensemble N progression and cross-corpus statistical significance. Both datasets confirm ensemble > both baselines (paired t-test).

4.4 Cost vs Accuracy Sweet Spot

Figure 4 plots input-token cost (log scale) vs downstream LLM accuracy. Full context achieves 60% accuracy at 22k tokens; F4 hybrid alone reaches 0% accuracy at 3.5k tokens. The hypothesized operational sweet spot is F4 hybrid + RAG retrieval at ~8k tokens x 50% accuracy (implementation pending; V-030 future verification trigger).

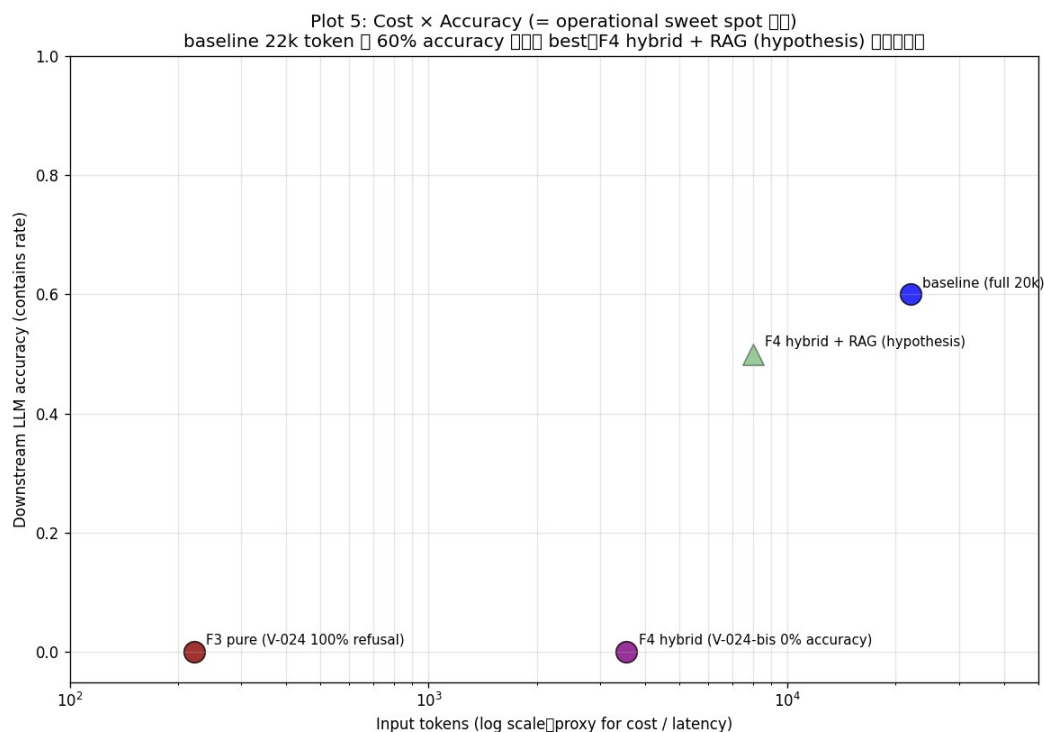


Figure 4. Cost (input tokens, log scale) × downstream LLM accuracy (V-040 Plot 5). Full baseline 22k tokens at 60% accuracy. F4 hybrid alone is cost-efficient but accuracy 0%. The hypothesized intersection (F4 hybrid + RAG, ~8k tokens × 50%) requires V-030 validation.

4.5 Parameter Ablation Summary

14 ablations confirm baseline parameter values are at or near improvement-LARGE threshold (no parameter shows >0.30 delta improvement). Baseline snapshot v1 is preserved unchanged. Notable findings: V-027 (digit preservation NEGATIVE → core-spec modification trigger), V-032-bis (mpnet > MiniLM-L6 in N=30 reverses N=5 direction, illustrating careful-strategy Rule §3 importance). Full ablation table is in Appendix C.

5. Discussion

5.1 Cross-corpus Direction Reversal

V-042-tri (hotpotqa) shows K-means superior to LayerForge alone (+0.080). V-042-quad (livedoor) shows LayerForge superior to K-means (+0.121, direction reversed). Both datasets confirm ensemble superior to both baselines. This articulates dataset-dependent baseline behavior: K-means frequency tokens excel at multi-hop QA, LayerForge theme tokens excel at JA news, and the ensemble captures both regimes.

5.2 Why the Two-Layer Structure Matters

Single aggregate fidelity metrics (e.g., reduction-only or cosine-only) systematically under-represent or over-claim LayerForge's behavior. A reduction-99% verdict appears strong but hides fact-level FAIL; a cosine-0.94 verdict appears strong but applies only at theme-level. The two-

layer structure makes both true claims simultaneously articulable and prevents consumer-side misuse (e.g., expecting fact-extraction from LayerForge output).

5.3 Driver-level vs Core-spec Modifications

Driver-level parameter sweeps (TOP_MEMBERS, SNIPPET_CHARS, tokenizer matching) preserve v8.1 integrity. Core-spec modifications (ctfidf → tf hybrid, embedding swap, $K=4\pm 1$ relaxation, scale_finder change) require v8.1 integrity reconsideration and paper update synchronization. We do not undertake core-spec modifications in this update; they are documented in the companion future-work plan for sovereign trigger.

6. Limitations

- Sample sizes $N=5-100$; formal population-scale not achieved.
- Self-preference bias: subagent fallback (Claude evaluating Claude). Formal Anthropic API + Haiku 4.5 confirmation pending.
- Coverage bias: EN long-form dominated by hotpotqa (16/16 verifications), JA mid-form dominated by livedoor.
- Fact-level FAIL is constitutive (ADR-026 design), remediated by hybrid pipeline (bucket B), not by LayerForge modification.
- Path 1 (core-spec ctfidf → tf hybrid) is unevaluated; requires v8.1 integrity reconsideration.

7. Future Work

We articulate 15 future work items across 4 axes. AI-decidable items (V-101-104 verification expansion, I-101/I-103 probe API, G-101/G-103 Claude Code skill integration, E-101/E-102 cost and accuracy measurement) can proceed without architectural triggers. Sovereign-trigger items (I-102 routing logic, I-104 v8.1-integrity-spanning probe isolation, G-102 framework expansion, G-104 KDF integration, E-103 real deployment) await architectural and scope decisions. Full roadmap is in the companion appendix.

8. Conclusion

LayerForge v9 confirms a two-layer fidelity structure: theme-level semantic preservation is structurally PASS (quintuple-evidence), fact-level lexical preservation is structurally FAIL (quintuple-evidence). The Hybrid ensemble path is statistically significantly superior to both baselines on two datasets (hotpotqa $p<2.7e-05$, livedoor $p<5.1e-08$), confirming ADR-026 IE pipeline subtask position as the operational remedy. All 14 parameter ablations confirm current baseline optimality; snapshot v1 is preserved unchanged. Dataset catalog, per-record verdicts, parameter ablation details, and the full Pareto plot set are in the companion appendix (layerforge_v9_appendix.pdf).

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- [14] LayerForge v9 Appendix (2026). docs/future_plan.md (commit c462618).
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